Local actions in rank two over a projective line

Human Compression project

We reprove the principal classification in Part I of De Saedeleer and Leemans, *On the rank two geometries of the groups* $\mathrm{PSL}(2,q)$, Ars Mathematica Contemporanea 3 (2010), 177–192. Throughout, $q \geq 4$ is a prime power, $d = \gcd(2, q-1)$ and $G = \mathrm{PSL}(2,q)$. The small nonsimple groups are not included in this standing hypothesis. We retain Dickson's classification of the proper subgroups of G, including the usual maximal-subgroup and natural-subfield clauses. The projective-line action, torus normalizers and the orders of PSL and PGL are classical inputs. We do not retain the source's principal geometry tables as premises.

The reduction below uses two elementary facts about primitive actions. It leaves two uniform families and fourteen small fields. Exact subgroup intersections and graph computations then determine every entry of the finite list. We distinguish ordered parabolics from the unordered convention stated on page 181 of the source. Two conjugacy counts differ between those conventions. The corrections to counts and diagram labels are recorded explicitly at the end.

1 The local action test

For distinct proper subgroups $H, K < G$, let $B = H \cap K$. Vertices of the two types are the right cosets of H and K; incidence means nonempty intersection. The neighbours of the base H-vertex are indexed by H/B, and similarly at the K-vertex. Every incident pair is a translate of the base pair. Thus the geometry is flag-transitive, and is connected precisely when $\langle H, K\rangle = G$.

Lemma 1. *The rank-two system is locally doubly transitive and residually weakly primitive if and only if at least one of H, K is maximal in G and*

$$|B\backslash H/B| = |B\backslash K/B| = 2.$$

These conditions imply firmness and connectedness. For fixed H, conjugacy classes of ordered systems are the $N_G(H)$-orbits on the eligible K; the analogous $\mathrm{Aut}(G)$-classes use $\mathrm{Aut}(G)_H$. For unordered systems, one must additionally identify the orbits transported by interchanging H, K.

Proof. The orbits of a point stabilizer B on H/B are precisely its double cosets. Exactly two orbits means that the nonbase points form one orbit, so the action is doubly transitive and its degree is at least two. It is primitive, whence B is maximal in each parabolic. Conversely these local actions give every nonempty-residue condition; the empty-residue condition is that one parabolic be maximal in G. The other parabolic is not contained in it, since their intersection has proper index in both. They therefore generate G. The orbit statements follow by normalizing the first subgroup. If a transporter sends K to H, its image of H gives the additional identification. Two choices of transporter differ by the indicated stabilizer. □

Lemma 2. *In a faithful primitive action, a nontrivial abelian normal subgroup is regular. A doubly transitive action with cyclic regular normal subgroup has prime degree; if its point stabilizer has order at most two, that degree is two or three. In every doubly transitive action of L on L/B,*

$$[L : B] - 1 \;\mid\; |B|.$$

Proof. The orbits of a normal subgroup form blocks, so a nontrivial normal subgroup in a faithful primitive action is transitive. If it is abelian, all its point stabilizers coincide and lie in the kernel, proving regularity. For a cyclic regular normal subgroup, any proper nontrivial characteristic subgroup would again be normal and transitive, which is impossible. Its order is therefore prime. The point stabilizer is transitive on all other points, proving both the bound on the degree and the divisibility assertion. □

2 The two uniform families

Write $P = V : C_t$ for a Borel subgroup, where $V = (\mathbb{F}_q, +)$ and $t = (q-1)/d$. A generator of its scalar complement generates the field $\mathbb{F}_q$: indeed $t > \sqrt{q} - 1$, whereas any proper subfield has multiplicative order at most $\sqrt{q} - 1$. Consequently its only invariant additive subgroups are $0, V$.

Theorem 3. *If one parabolic is a Borel subgroup, then q is even. There are exactly two systems up to conjugacy, hence also up to* $\mathrm{Aut}(G)$*:*

1. *two copies of the projective line, with unequal points incident;*
2. *the points and unordered pairs of points of the projective line, with incidence given by containment.*

Their local arc-transitivity depths are respectively two and three. Their type sizes, local degrees, gonalities and type diameters are

	type sizes	*local degrees*	*gonality*	*diameters*
1	$(q+1, q+1)$	(q, q)	2	$(3, 3)$
2	$(q+1, q(q+1)/2)$	$(q, 2)$	3	$(3, 4)$.

The second is a partial linear space; the first is not. Both are residually primitive. The first is the usual bipartite double of the complete graph underlying the second.

Proof. In a primitive coset action of P, the kernel intersected with V is scalar-invariant. If the image of V is trivial, the action factors through the cyclic complement and double transitivity forces degree two. Thus the point stabilizer contains V. No proper subgroup of G outside this Borel contains V: the subgroup alternatives show that a proper subfield group has a strictly smaller characteristic Sylow subgroup, while the small exceptional groups give no exception for $q \geq 4$. In the remaining affine case the unique fixed projective point forces the same Borel. Hence a second parabolic is impossible in this case.

Otherwise V acts faithfully and regularly, by Lemma 2, and the degree is q. Double transitivity requires the scalar complement to be transitive on the nonzero translations. It follows that $t = q - 1$, so q is even. The point stabilizer is a complement C_{q-1}: a generator of its projection is an affine map with a unique fixed point, so translation conjugacy puts the whole complement in scalar form.

A cyclic torus C_{q-1} fixes two projective points. A proper overgroup acting doubly transitively on its cosets is one of their two Borels or its dihedral normalizer. To check this directly from Dickson, an affine overgroup has regular translation group and a scalar complement of order one less than its size, forcing the full Borel. A cyclic subgroup cannot be maximal in a genuine subfield PSL or PGL group except in the small S_3, A_4 cases. The exceptional groups likewise supply only A_4 with stabilizer C_3, which is the Borel at $q = 4$. A dihedral overgroup normalizes the torus and is therefore its full normalizer. This proves completeness.

The two stabilizer configurations give exactly the stated incidence models. The projective action is doubly transitive, so they each form one conjugacy class. For even q, $G = \mathrm{PGL}(2, q)$ acts sharply transitively on ordered triples of distinct projective points: the fractional linear

transformation carrying a given triple to $(\infty, 0, 1)$ is unique. In the first graph, two-arcs starting at a fixed point correspond to distinct triples. At length three the last point may equal the first underlying point or not, so these arcs cannot form one orbit. In the point-pair incidence graph, arcs alternate between points and edges. Sharp transitivity on distinct triples gives local transitivity through length three. Starting at an edge, length-four arcs distinguish return to its other endpoint from a new endpoint, so the depth is exactly three. Direct paths in the two incidence models give the displayed gonalities and diameters. Two vertices in the first part of model 1 have $q - 1 \geq 3$ common neighbours, excluding the partial-linear property; it is not a digon either. Model 2 is the incidence geometry of a complete graph. The classical maximality of these Borels and split torus normalizers proves residual primitivity. □

3 Why only small exceptional fields remain

Let $H = D_{2n}$ be a full torus normalizer, where $n = (q \pm 1)/d$.

Lemma 4. *If H acts doubly transitively on H/B, then either $B = C_n$, or $B = D_{2t}$ with $n = 2t$ or $n = 3t$. Apart from the uniform families in Theorem 3, every eligible system occurs for*

$$q \in \{4, 5, 7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 29, 31\}.$$

Proof. Pass to the faithful quotient of the primitive action. The cyclic rotation subgroup either has trivial image, giving the two-point quotient with $B = C_n$, or is regular of prime order. The image of the point stabilizer then has order at most two. Lemma 2 gives degrees two or three, and the remaining subgroups are exactly the indicated dihedral ones.

First let $B = C_n$, with $n > 2$. Any dihedral overgroup normalizes this semisimple cyclic subgroup and lies in $N_G(B) = H$, so cannot be the second parabolic. Cyclic overgroups likewise give none. Suppose the second parabolic is affine, with normal elementary abelian characteristic subgroup of order p^e. Since $p \nmid n$, its image is regular with trivial kernel. The cyclic scalar stabilizer acts freely on its nonzero vectors and transitively on them, so $n = p^e - 1$, with $e \leq r$ when $q = p^r$. For odd p the split equation $2p^e = p^r + 1$ is impossible modulo p. The nonsplit equation is $2p^e = p^r + 3$; it forces $p = 3$ and then $e = r = 1$, outside $q \geq 4$. For even p only the split solution $e = r$ remains, giving the full Borel. In the exceptional groups a maximal cyclic point stabilizer can only be $C_3 < A_4$. In a subfield PSL or PGL group, a cyclic subgroup lies in a proper torus normalizer unless it belongs to a Borel; that Borel is cyclic only in the small S_3, A_4 cases. These are already covered, or give $n = 3$. The cases $n \leq 3$ have $q \leq 7$ and are included in the finite list.

Now suppose $B = D_{2t}$ and $t > 2$. Its characteristic rotation subgroup C_t is semisimple. A second dihedral parabolic would normalize C_t, and again lie in H, which is impossible. If a second affine parabolic has normal characteristic subgroup V, then $V \cap B = 1$. In odd characteristic this follows from orders. In characteristic two, t is odd and D_{2t} has no nontrivial normal 2-subgroup. The regular normal image of V would therefore identify B with a cyclic complement, contrary to $t > 2$.

Consider a second subfield group $L = \mathrm{PSL}(2, s)$ or $\mathrm{PGL}(2, s)$. Because B is maximal in L, the classical subgroup list makes it a full torus normalizer, unless L is one of the small S_3, A_4, S_4, A_5 cases. Indeed a dihedral Borel is possible only at these small fields, and a dihedral subfield subgroup itself reduces to S_3. In odd PSL, $|B| \leq s + 1$ and $[L : B] \geq s(s - 1)/2$, so Lemma 2 forces $s \leq 4$. In even PSL or in PGL, the corresponding bounds are $|B| \leq 2(s + 1)$ and $[L : B] \geq s(s - 1)/2$, forcing $s \leq 6$. The only additional possibility is $\mathrm{PGL}(2, 5)$: its torus normalizers have orders $8, 12$ and indices $15, 10$, respectively, and neither $14 \mid 8$ nor $9 \mid 12$. It supplies no case.

The dihedral doubly transitive point stabilizers in the remaining exceptional groups are $S_3, D_8 < S_4$ and $D_{10} < A_5$. This follows from their elementary subgroup lists and the same

divisibility test; A_4 supplies only its cyclic C_3. Thus $t \in \{3,4,5\}$ and $n \in \{2t, 3t\}$. Solving $q \pm 1 = dn$ among prime powers gives the displayed finite fields. Finally $t \leq 2$ gives $n \leq 6$ and adds only $q \leq 13$ from that list. We include the nonmaximal torus cases $q = 5, 7, 9, 11$ explicitly, rather than assuming that the specified torus parabolic is the maximal one. □

4 The finite list and its verification

Table 1 removes repetitions of the two uniform families and the isomorphic ambient groups $\mathrm{PSL}(2,4) \cong \mathrm{PSL}(2,5)$. Here c counts ordered conjugacy classes; u counts unordered conjugacy classes. Every row is a single $\mathrm{Aut}(G)$-class. Type sizes and degrees follow from $|G|/|H|, |G|/|K|$ and $|H|/|B|, |K|/|B|$. The last columns give gonality, the two type diameters, and local arc-transitivity depth.

q	H	K	B	c	u	$(g; d_0, d_1)$	s
4	S_3	C_2^2	C_2	1	1	$(5; 5, 6)$	3
4	S_3	S_3	C_2	1	1	$(3; 5, 5)$	2
11	D_{12}	D_{12}	C_2^2	2	1	$(5; 7, 7)$	3
13	D_{12}	D_{12}	C_2^2	2	1	$(6; 9, 9)$	3
16	D_{30}	A_5	D_{10}	1	1	$(4; 7, 7)$	3
17	D_{16}	S_4	D_8	2	2	$(9; 14, 14)$	7
17	D_{18}	S_4	S_3	2	2	$(4; 8, 8)$	3
19	D_{20}	A_5	D_{10}	2	2	$(5; 8, 7)$	3
23	D_{24}	S_4	D_8	2	2	$(6; 11, 11)$	3
25	D_{24}	S_4	D_8	2	2	$(5; 13, 13)$	3
29	D_{30}	A_5	D_{10}	2	2	$(4; 8, 7)$	3
31	D_{30}	A_5	D_{10}	2	2	$(4; 8, 7)$	3

Table 1: The exceptional systems. The two $q = 4$ rows also occur at $q = 5$; the A_4, S_3, C_3 row at $q = 5$ is the $q = 4$ uniform point-pair geometry.

Theorem 5. *For $q \geq 4$, Theorem 3 and Table 1 give all locally doubly transitive RWPRI rank-two systems having a Borel or full torus normalizer as one parabolic. The finite rows are partial linear spaces. Both parabolics are maximal except in the $q = 4$, S_3, C_2^2 row and the $q = 25$ row; precisely the other finite rows are residually primitive.*

Proof. The two preceding results reduce completeness to the fourteen fields. The certificate constructs G in its natural permutation action and lists all its subgroups from conjugacy-class representatives and their conjugates. For each representative full torus normalizer H, it tests every proper K by literal intersection, the two double-coset counts, and actual maximality of at least one parabolic. No abstract order match substitutes for the intersection or action test. Lemma 1 proves that precisely the eligible systems survive. The normalizer and automorphism-stabilizer orbits give their ordered counts. Actual transporters implement the unordered identifications. The full returned inventory is bound to the certificate, including the empty cases and the uniform rows removed from the display.

For each representative, right cosets of B supply exactly $|G : B|$ edges between its two parabolic coset sets. The permutation action on the graph is faithful. Breadth-first distances from a vertex of each type give the diameters, since G is transitive on each type. The girth is computed both by deleting an edge and finding the distance between its endpoints, and by an independent all-root breadth-first computation. All finite rows have girth at least six, proving the partial-linear property.

For completeness, local arc-depth is computed as follows. Start at a vertex and its stabilizer. If the current pointwise path stabilizer is transitive on the neighbours of the last vertex other

than its predecessor, extend by one chosen neighbour and intersect with its stabilizer. Otherwise stop. Inductively all arcs of the attained length form one orbit, so this test is necessary and sufficient at the next length. Repeated vertices other than immediate backtracking are allowed, as required by the source definition. Perform this from both types and take the smaller depth. The exact outputs are those in the table. □

5 A scalar-subspace count

The source's Remark 4.3 gives eleven conjugacy classes of elementary abelian subgroups of order sixteen in $\mathrm{PSL}(2,64)$, in correction of an earlier enumeration attributed there to Dickson. This count has a short independent proof. A nontrivial unipotent subgroup fixes a unique projective point. After fixing that point, subgroups of order sixteen are the four-dimensional $\mathbb{F}_2$-subspaces T of $\mathbb{F}_{64}$, with scalar multiplication by $\mathbb{F}_{64}^{\times}$ giving precisely conjugacy. Translations centralize them.

The set $E = \{a \in \mathbb{F}_{64} : aT \subseteq T\}$ is a subfield: it is closed under addition and multiplication, and a nonzero scalar acts bijectively on the finite space T, so its inverse also preserves T. Its degree over $\mathbb{F}_2$ divides both six and four. Thus $E = \mathbb{F}_2$ or $\mathbb{F}_4$. There are $\binom{6}{4}_2 = 651$ such subspaces in all, and $\binom{3}{2}_4 = 21$ are $\mathbb{F}_4$-spaces. The latter have scalar stabilizer of order three, so form one orbit of length $63/3 = 21$. The remaining 630 subspaces have trivial scalar stabilizer and form ten orbits of length 63. Their normalizers in G have orders $64 \cdot 3$ and 64, giving one subgroup conjugacy class of length 1365 and ten of length 4095, exactly as stated in the source. This argument counts subspaces and their scalar orbits separately; treating the 651 subspaces as conjugacy classes would conflate the two sets.

6 Scope and corrections

The first four sections reprove the main existence and completeness result, the local arc-depth column, and the diagram invariants of the source's Theorem 1.1 and Propositions 5.3, 5.8 and 5.12. The omitted intermediate overgroup proofs are supplied by the two uniform action arguments and Lemma 4. Dickson's classification and its maximal-subgroup clauses remain attributed classical inputs. The names Petersen, Desargues and Perkel refer to earlier identifications; no priority claim about these graphs is made here.

The source defines conjugacy using the unordered set of parabolics. At $q = 11, 13$, its two ordered D_{12}, D_{12} classes are exchanged by type interchange. Thus the printed count two is correct for ordered parabolics, and the count under its stated unordered convention is one. The certificate records an actual transporter and checks the distinct ordered orbits. The $\mathrm{Aut}(G)$ count remains one.

The split torus normalizer has index $q(q+1)/2$, as printed correctly on page 186; the repeated $q(q-1)/2$ on page 188 is incorrect. In the odd-characteristic Table 3, the nonsplit normalizer index is $q(q-1)/2$, with a factor two missing from the printed denominator. In the $q = 17$ diagram on page 189, the D_{16} type has $2448/16 = 153$ vertices, not 136. All other displayed exceptional distances, gonalities and local arc-depths agree with the exact computations. The unqualified prime-order reflection clause in Table 4 is replaced by Lemma 4: D_{2p} on reflection cosets is doubly transitive only for $p = 3$ (with the degree-two boundary treated separately).

The simple-group hypothesis matters at the boundary. In $\mathrm{PSL}(2,2) = S_3$, two distinct order-two parabolics give the six-cycle. The two uniform rows coincide there, and this cycle is locally arc-transitive at every length: after its first edge a nonbacktracking walk has a unique continuation, and the vertex stabilizer exchanges the two choices of direction. In $\mathrm{PSL}(2,3) = A_4$ no eligible system exists. A cyclic order-three parabolic has no doubly transitive nontrivial coset action. A Klein-four parabolic would require an order-two intersection, but every proper

overgroup of that intersection lies in the same unique Klein four. This also settles the boundary if the displayed source theorem is read without its surrounding simple-group convention.

Classical subgroup classifications remain borrowed inputs; the finite certificate supplements the written reduction and is not a proof of those classifications.